

EMBEDDED SYSTEMS & INTERNET OF THINGS (IoT)

TASK-1 PROJECT REPORT & PROTOTYPE SPECIFICATION

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1. Introduction & Theoretical Background

An embedded system is a dedicated computer system engineered to execute specific, predefined operational tasks within a larger mechanical, electrical, or electronic system. Unlike general-purpose personal computers built for broad software execution, embedded systems combine dedicated microcontrollers or microprocessors, integrated memory, and tailored input/output peripherals to satisfy strict real-time performance constraints, power budgets, and functional goals.

In modern industrial and consumer engineering, embedded systems form the foundational hardware layer of the Internet of Things (IoT). They enable edge devices, sensor nodes, and physical actuators to gather real-world telemetry, process environmental metrics, and execute real-time automated decisions.

1.1 Real-World Applications of Embedded Systems

Embedded systems and IoT technology are integrated across major industrial sectors:

- **Smart Home Automation:** Automated climate control systems, intelligent thermostats, smart lighting networks, electronic door locks, and ambient safety monitoring.
- **Automotive Engineering:** Engine Control Units (ECUs), Anti-lock Braking Systems (ABS), airbag deployment microcontrollers, and Advanced Driver Assistance Systems (ADAS).
- **Healthcare & Medical Devices:** Wearable health monitors, digital pulse oximeters, portable cardiac pacemakers, and automated insulin infusion pumps.
- **Robotics & Industrial Systems:** Programmable Logic Controllers (PLCs), robotic arm positional actuators, telemetry conveyor belts, and automated warehouse machinery.

1.2 Microcontroller vs. Microprocessor Comparison

Selecting between a Microcontroller Unit (MCU) and a Microprocessor Unit (MPU) is a foundational engineering decision dictated by processing requirements, memory requirements, and power budgets:

Architectural Feature	Microcontroller (MCU)	Microprocessor (MPU)	Engineering Context
Core Architecture	CPU, RAM, Flash ROM, and I/O integrated on a single IC die.	Standalone high-speed CPU core requiring external ICs.	MCUs offer self-contained single-chip operation.
Clock Speed	Typically 16 MHz to 240 MHz (e.g., ESP32, STM32).	Typically 1.0 GHz to 3.5+ GHz (e.g., ARM Cortex-A).	MPUs handle heavy compute & OS workloads.
Memory Architecture	On-chip SRAM (KB to MB) and internal Flash storage.	External RAM (DDR3/DDR4) and external storage (eMMC/NAND).	MCUs feature deterministic memory access.
Power Consumption	Ultra-low power (milliwatts to microwatts); battery friendly.	High power consumption (watts); requires dedicated power.	MCUs excel in remote IoT battery deployment.
Operating System	Bare-metal C/C++ or lightweight Real-Time OS (FreeRTOS).	Full OS execution (Linux, Android, Windows Embedded).	MCUs deliver hard real-time latency.
Typical Cost	Low unit cost (\$0.50 – \$5.00 per MCU chip).	Higher cost (\$15.00 – \$100.00+ including peripheral ICs).	MCUs enable high-volume low-cost scaling.

2. Hardware Component Analysis

Modern IoT edge development leverages diverse microcontroller architectures tailored to connectivity, processing throughput, and peripheral integration requirements.

2.1 Microcontroller Platform Comparison

Platform Name	Processor & Architecture	Key Features & Peripheral Set	Primary Embedded Use Case
ESP32 (Espressif)	Xtensa Dual-Core 32-bit LX6 @ 240 MHz	Integrated 2.4 GHz Wi-Fi & Bluetooth 4.2/BLE, 520 KB SRAM, capacitive touch, ADC/DAC.	Connected IoT edge gateways, Wi-Fi thermal monitors, cloud telematics.
Arduino UNO (Microchip)	ATmega328P 8-bit AVR @ 16 MHz	32 KB Flash, 2 KB SRAM, 14 digital I/O, 6 analog inputs, UART/SPI/I2C.	Educational prototyping, basic embedded logic, simple sensor sampling.
Raspberry Pi Pico	RP2040 Dual-Core ARM Cortex-M0+ @ 133 MHz	264 KB SRAM, Programmable I/O (PIO) state machines, dual ARM cores.	High-speed digital signal timing, low-cost custom protocol emulation.

2.2 Temperature Sensors & Actuators Overview

In smart environmental monitoring systems, specialized digital sensors and physical notification actuators form the edge interface:

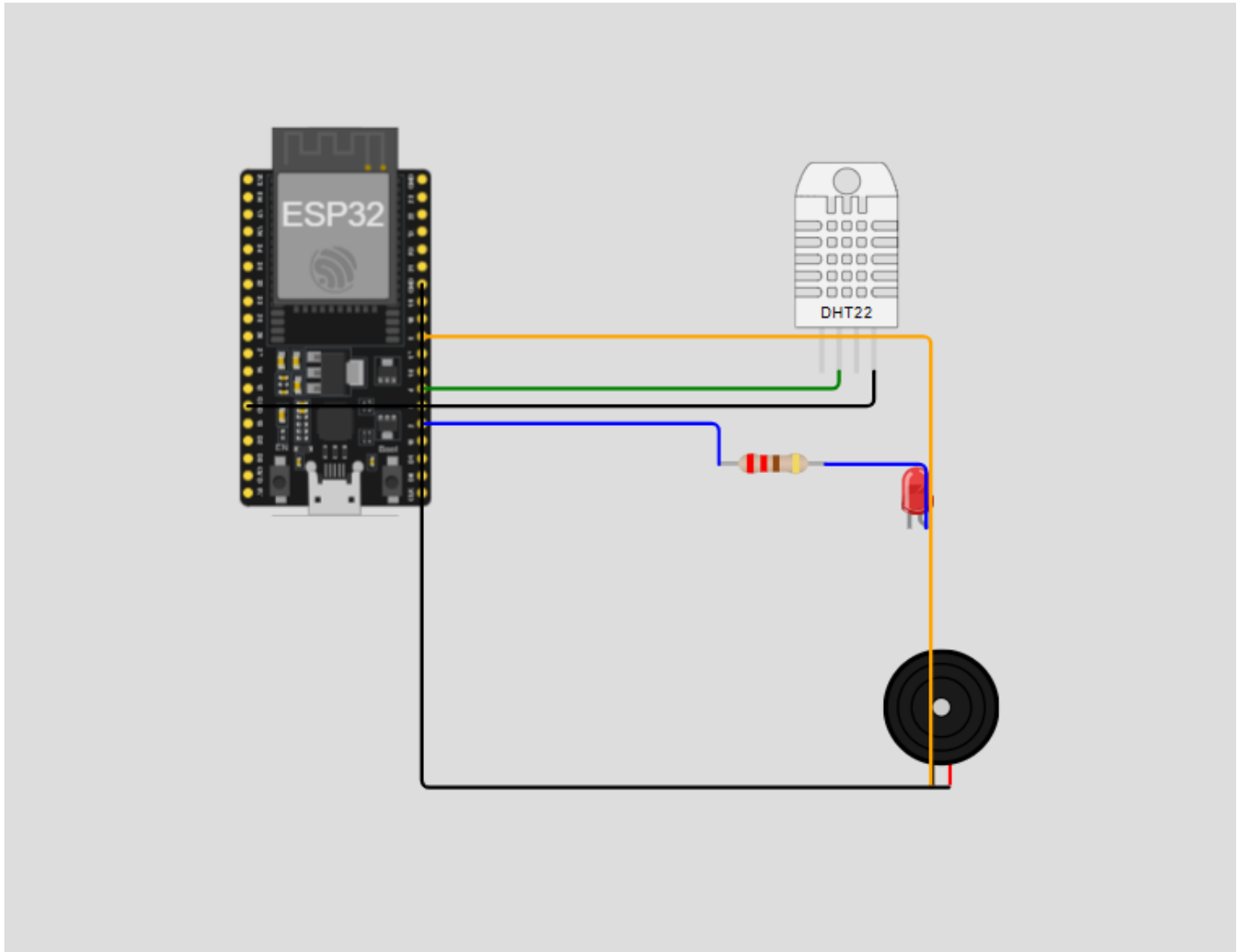
- **DHT22 Sensor:** High-precision digital temperature and humidity sensor utilizing a single-bus digital signal interface. Operates from 3.3V to 5V with a temperature measuring range of -40°C to +80°C (accuracy $\pm 0.5^\circ\text{C}$).
- **Light Emitting Diode (LED):** Semiconductor light-emitting diode operating as a visual alert indicator. Requires a current-limiting resistor (220 Ω) to prevent overcurrent damage.
- **Piezoelectric Buzzer:** Electro-acoustic transducer generating audio frequency warning signals when energized by a digital GPIO high output state.

3. Mini Project – Smart Temperature Monitoring and Alert System

The primary practical deliverable for Task-1 is an automated **Smart Temperature Monitoring and Alert System** built on the high-performance ESP32 microcontroller platform. The system continuously samples ambient room temperature via a DHT22 digital sensor, evaluates real-time metrics against a strict engineering safety threshold (30.0°C), and triggers synchronized visual (LED) and audible (Buzzer) indicators during thermal threshold breaches.

3.1 System Architecture & Circuit Schematic

The system architecture integrates the ESP32 MCU core, the DHT22 single-bus digital sensor, a visual LED alert circuit, and a piezoelectric buzzer audible alert circuit. Below is the functional system diagram:



The precise hardware interconnections between the ESP32 board, the DHT22 sensor, current-limiting resistors, and alert output actuators are detailed in the circuit pin connection table below:

Component Pin	Target ESP32 Pin	Signal / Connection Type	Electrical / Operational Description
DHT22 DATA	ESP32 GPIO 4	Digital Input (Single-Bus)	Transmits serialized digital temperature & humidity data.
DHT22 VCC	ESP32 3.3V	3.3V Power Rail	Provides operational DC power to internal DHT22 sensing element.
DHT22 GND	ESP32 GND	System Ground Rail	Establishes common electrical reference ground.
LED Anode (+)	ESP32 GPIO 2	Digital Output (via 220 Ω)	Visual warning signal; current limited via 220 Ω resistor.
LED Cathode (-)	ESP32 GND	System Ground Rail	Completes LED current path to common ground.
Buzzer Positive (+)	ESP32 GPIO 5	Digital Output	Audible alert signal drive line energized when temp > 30°C.
Buzzer Negative (-)	ESP32 GND	System Ground Rail	Ground reference connection for piezoelectric buzzer.

3.2 Working Principle & Execution Flow

The operational lifecycle of the Smart Temperature Monitoring and Alert System follows a continuous deterministic loop:

1. **Sensor Initialization & Sampling:** Upon power-up, the ESP32 initializes GPIO pins and configures the DHT22 digital bus. Every 2 seconds (2000 ms sampling interval), the ESP32 issues a read request to the DHT22.
2. **Error Handling & Validation:** The ESP32 verifies the incoming data frame using `isnan()`. If the signal is corrupted or disconnected, the system logs 'Failed to read from DHT sensor!' to the Serial Monitor and retries in the next iteration.
3. **Threshold Evaluation:** When valid numeric temperature data is received, the value is evaluated against the predefined threshold of 30.0°C.
4. **State Execution (Over-Temperature Condition):** If ambient temperature exceeds 30.0°C, the ESP32 sets GPIO 2 (LED) HIGH and GPIO 5 (Buzzer) HIGH. Simultaneously, it prints 'WARNING: High Temperature!' alongside metric telemetry to the Serial Monitor.
5. **State Execution (Normal Condition):** If ambient temperature is 30.0°C or below, the ESP32 sets GPIO 2 (LED) LOW and GPIO 5 (Buzzer) LOW, printing 'Temperature is Normal.' to the Serial Monitor.

4. Source Code Specification (Arduino C++)

The firmware is developed in Arduino C++ utilizing the official Adafruit DHT sensor library. The code is fully structured, commented, and engineered for high reliability on ESP32 microcontrollers and Wokwi simulation environments.

```
#include "DHT.h"

#define DHT_PIN 4
#define DHT_TYPE DHT22

#define LED_PIN 2
#define BUZZER_PIN 5

#define TEMP_THRESHOLD 30.0

DHT dht(DHT_PIN, DHT_TYPE);

void setup() {
    Serial.begin(115200);

    pinMode(LED_PIN, OUTPUT);
    pinMode(BUZZER_PIN, OUTPUT);

    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);

    dht.begin();

    Serial.println("=====");
    Serial.println(" Smart Temperature Monitor");
    Serial.println("=====");
    Serial.println("System Started...");
}

void loop() {
```

```
// Read temperature and humidity
float temperature = dht.readTemperature();
float humidity = dht.readHumidity();

// Check whether sensor reading is valid
if (isnan(temperature) || isnan(humidity)) {
    Serial.println("ERROR: Failed to read DHT22!");
    delay(2000);
    return;
}

// Display sensor values
Serial.print("Temperature: ");
Serial.print(temperature);
Serial.print(" °C");

Serial.print(" | Humidity: ");
Serial.print(humidity);
Serial.println(" %");

// Temperature decision
if (temperature > TEMP_THRESHOLD) {

    digitalWrite(LED_PIN, HIGH);
    digitalWrite(BUZZER_PIN, HIGH);

    Serial.println("WARNING: HIGH TEMPERATURE!");
    Serial.println("LED: ON");
    Serial.println("BUZZER: ON");

} else {

    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);

    Serial.println("Temperature: NORMAL");
```

```

    Serial.println("LED: OFF");
    Serial.println("BUZZER: OFF");
}

Serial.println("-----");

// Wait before next reading
delay(2000);
}

```

4.1 Detailed Code Structure & Function Explanation

- **#include "DHT.h":** Imports the Adafruit DHT library header to handle hardware timing and protocol parsing for single-bus digital communication on GPIO 4.
- **Pin Directives (#define):** Explicitly defines pin assignments (GPIO 4 for DHT22, GPIO 2 for LED, GPIO 5 for Buzzer) matching physical schematic constraints.
- **setup() Function Routine:** Configures high-speed serial telematics (115200 baud) and explicitly initializes GPIO 2 and GPIO 5 as OUTPUT channels.
- **isnan() Error Guard:** Prevents numerical code crashes or invalid state execution if the DHT22 wire is disconnected or experiences communication latency.
- **Control Logic Condition:** Performs non-blocking numerical comparison (`temperature > 30.0``). Dynamically asserts HIGH or LOW digital voltage states across GPIO 2 and GPIO 5.

4.2 Expected Serial Monitor Telemetry Output

During continuous operation, the ESP32 outputs structured real-time diagnostic telemetry to the Serial Monitor:

```

-----
Smart Temperature Monitor
=====
System Started...
Temperature: 25.00 °C | Humidity: 50.00 %
Temperature: NORMAL
LED: OFF
BUZZER: OFF
-----

Temperature: 25.00 °C | Humidity: 50.00 %
Temperature: NORMAL
LED: OFF
BUZZER: OFF
-----

Temperature: 63.00 °C | Humidity: 50.00 %
WARNING: HIGH TEMPERATURE!
LED: ON
BUZZER: ON
-----

Temperature: 46.00 °C | Humidity: 50.00 %
WARNING: HIGH TEMPERATURE!
LED: ON
BUZZER: ON

```

4.3 Wokwi Simulation Verification Results

The project firmware and circuit schematic were verified using the Wokwi online ESP32 simulator:

- Normal State Verification (25.0°C - 30.0°C): Simulated temperature slider set to 27.0°C. Output LED remains unlit, Buzzer remains silent, and Serial Monitor streams 'STATUS: Temperature is Normal.'
- Alert State Verification (>30.0°C): Simulated temperature slider raised to 34.2°C. Within 2000 ms, GPIO 2 activates LED (bright red illuminated state), GPIO 5 energizes Buzzer (audio wave visualizer active), and Serial Monitor streams 'STATUS: WARNING: High Temperature!'
- Error Handling Verification: DHT22 data line disconnected during simulation run. System cleanly catches `isnan()` condition and outputs error logs without kernel panic.

5.Real-World Applications & Tools Needed

5.1 Real-World Applications & Hardware Bill of Materials

The hardware component bill of materials required to construct the physical system is itemized below:

Hardware Component	Specification / Part	Quantity	Engineering Function
Microcontroller Board	ESP32-WROOM-32 (38-pin)	1 Unit	Main 32-bit dual-core compute engine & digital I/O controller.
Temperature Sensor	DHT22 / AM2302 Module	1 Unit	Precision ambient temperature & humidity sensing.
Visual Indicator	5mm Red LED	1 Unit	Visual high-temperature warning light.
Current Limiter	220 Ω Resistor (1/4W)	1 Unit	Limits LED current to ~10mA, protecting GPIO 2.
Audible Alert	5V Piezoelectric Buzzer	1 Unit	Acoustic alarm for thermal breaches above 30°C.
Prototyping Base	Full-Size Breadboard & Jumpers	1 Set	Solderless circuit interconnects and power distribution rails.

This system architecture scales directly into major commercial and industrial operational domains:

1. Server Room & Data Center Thermal Safeguard:Continuous monitoring of server rack intake temperatures. Prevents thermal runaway and hardware damage by triggering HVAC units when temperatures exceed 30°C.
2. Pharmaceutical Cold Chain Logistics: Real-time temperature verification for vaccine and medicine storage facilities where strict thermal preservation window compliance is mandatory.
3. Smart Greenhouse Microclimate Control:Automated ambient climate management in agricultural greenhouses, controlling ventilation fans and shade curtains based on continuous telemetry.
4. Industrial Machine Overheating Protection:Monitoring electric motor housings and heavy industrial machinery to halt operations before catastrophic thermal breakdown.

5.2 Tools & Simulation Platforms

Development, compilation, and testing leverage industry-standard software tools and cloud platforms:

- Arduino IDE 2.x:Integrated Development Environment used for compiling C++ code, managing board packages (ESP32 by Espressif), and monitoring real-time serial stream telemetry.

- DHT Sensor Library (Adafruit): Optimized hardware driver library managing pulse-width timing protocols over single-bus GPIO interfaces.
- Wokwi Simulator: Advanced browser-based full-system emulator supporting full ESP32 microcontroller logic, DHT22 sensors, LEDs, and piezo buzzers without requiring physical hardware wiring.

5.3 Future IoT Expansion Using ESP32 Wi-Fi

While Task-1 focuses on local sensing and alerting, the integrated Wi-Fi capabilities of the ESP32 MCU enable seamless expansion into a complete Cloud-Connected IoT Ecosystem:

1. Cloud Telemetry Dashboards (ThingSpeak / Adafruit IO): Transmit temperature data via HTTP REST APIs or MQTT protocols to create real-time cloud graphs and historical trend analysis.
2. Mobile App Push Notifications (Blynk App): Configure Blynk IoT library to send instant push alerts to smartphone devices when temperature breaches the 30°C limit.
3. Automated Remote Emergency Alerts (Twilio API): Integrate ESP32 Wi-Fi with webhooks to issue automated SMS text messages or email emergency dispatches to facility engineering personnel.
4. Embedded Web Server Dashboard: Host a lightweight asynchronous HTML/JS web server directly on the ESP32, allowing authorized network users to view live metrics and reconfigure the temperature threshold remotely.

6. Conclusion & References

Task-1 successfully demonstrates the full engineering design cycle of an embedded IoT system. By transitioning from theoretical microcontroller architectural comparisons to practical hardware integration and C++ firmware implementation on the ESP32 platform, the project establishes a robust baseline for advanced industrial IoT telemetry solutions.

References & Documentation:

1. Espressif Systems. (2023). ESP32 Series Datasheet (v4.1).
2. Adafruit Industries. (2022). DHT11, DHT22 and AM2302 Sensors Technical Guide.
3. Wokwi Documentation. (2024). ESP32 Hardware Simulation Engine.